

Data-Pruning Scores Break Under Distribution Shift

Supplementary Material

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S1 Additional critical-difference diagram

Figure S1 gives the Nemenyi critical-difference diagram at the most aggressive keep ratio, keep 10%, complementing the moderate-budget diagram in the main paper. Random selection is the single best-ranked method on both datasets, and every score trails it, which is the aggressive-budget end of the trend reported in the main text.



Fig. S1: Nemenyi critical-difference diagram of mean corruption-accuracy rank at the aggressive keep ratio of 10%, on CIFAR-10 (*left*) and CIFAR-100 (*right*); lower is better. Random selection is the single best-ranked method on both datasets, and every score trails it. Frozen final run.

S2 Accuracy by corruption severity

Figure S2 resolves accuracy by corruption severity at keep 50%. The gap between the DINOv2-feature coreset and random is present at the mildest severity and persists across the scale rather than opening up only at the harshest levels, consistent with the per-corruption breakdown in the main paper. The CLIP-feature coreset tracks random closely at every severity.

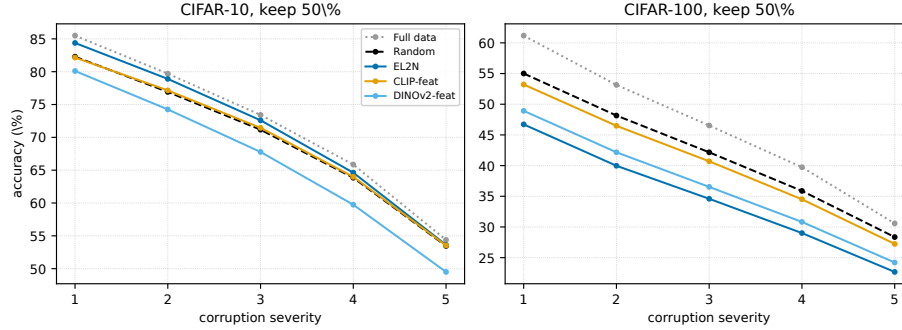


Fig. S2: Accuracy against corruption severity (1–5), averaged over the fifteen corruption types and five seeds, at keep 50%, on CIFAR-10 (*left*) and CIFAR-100 (*right*). Full-data and random references are dashed and dotted. The DINOv2-feature coresets trails random at every severity; CLIP-feature tracks it. Frozen final run.

S3 Clean-versus-corruption trade-off

Figure S3 places each coreset by its change against random on clean accuracy (horizontal) and on mean corruption accuracy (vertical). Points below the diagonal are less robust than their clean accuracy implies; points near zero horizontally but below zero vertically look like random on clean accuracy while being worse under shift. The DINOv2-feature coresets populate this region; the CLIP-feature coresets stay nearer the diagonal and the origin.

S4 Friedman omnibus

Table S1 gives the per-cell Friedman statistic and p -value behind the main-paper statement that score identity affects corruption accuracy at every operating point. All ten cells are significant at $p < 5 \times 10^{-4}$ and nine of the ten at $p < 10^{-5}$; the single least-extreme cell is CIFAR-10 at keep 90% ($\chi^2 = 25.0$, $p = 3.4 \times 10^{-4}$), as expected where only a tenth of the data is discarded, and the growth of χ^2 as the keep ratio falls tracks the widening method gap under more aggressive pruning.

S5 Class collapse and worst-class accuracy

Table S2 reports how far each score starves its thinnest class at keep 10% and the resulting worst-class corrupted accuracy at keep 30%. Keep-hardest scores retain only a handful of examples of the smallest class where random keeps many, and the worst-class corrupted accuracy falls in step.

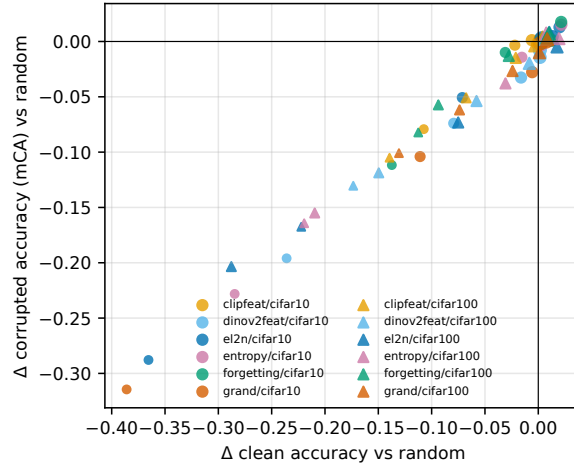


Fig. S3: Change against random on clean accuracy (horizontal) versus on mean corruption accuracy (vertical), one point per (score, dataset, keep ratio). Points below the diagonal are less robust than their clean accuracy implies; points near zero horizontally but below zero vertically look equivalent to random on clean accuracy while being worse under shift. Frozen final run.

S6 Compute budget

The frozen final run comprises 360 units and took 13.6 GPU-hours in total on a single NVIDIA A100 (40 GB), with a peak of 4.5 GB of GPU memory per unit; the self-contained scoring-choice ablation (110 units) added 4.0 GPU-hours on the same hardware. Wall-clock time and peak memory are recorded per unit in the released run manifest.

S7 Selection stability

Figure S4 shows the kept-set Jaccard overlap across the five scoring seeds. Score-based selection is far more stable than random, whose overlap follows the analytic baseline $\mathbb{E}[J] = r/(2 - r)$; the stably chosen subset is nonetheless the one that fails under shift.

S8 Asset attribution

The study builds on public assets: CIFAR-10 and CIFAR-100 [2], the common-corruption benchmark [1], the CLIP ViT-B/32 (OpenAI) and DINOv2 ViT-S/14 (LVD-142M) encoders [3, 4] used only as frozen feature extractors, and the timm library for the encoders. Software versions are pinned per unit (PyTorch 2.13, torchvision 0.28, timm 1.0.28). We use each asset under its released licence and add no new personal data or human-subject data.

Table S1: Friedman omnibus across the 15 corruption blocks at each keep-ratio ($k = 7$ methods, $\text{df} = 6$). All ten cells are significant at $p < 5 \times 10^{-4}$; nine of ten at $p < 10^{-5}$. χ^2 grows as the keep-ratio falls, tracking the widening method gap under more aggressive pruning.

Keep ratio	90%	70%	50%	30%	10%
CIFAR-10 χ^2	25.0	35.3	59.1	72.2	90.0
p	3.4×10^{-4}	3.8×10^{-6}	6.7×10^{-11}	1.5×10^{-13}	3.0×10^{-17}
CIFAR-100 χ^2	52.1	50.3	77.5	84.7	87.3
p	1.8×10^{-9}	4.0×10^{-9}	1.2×10^{-14}	3.8×10^{-16}	1.1×10^{-16}

Table S2: Mechanism. Left: mean smallest per-class retained count at keep 10% (how far a coreset starves its thinnest class). Right: worst-class corrupted accuracy at keep 30%. Keep-hardest scores collapse minority classes that random preserves. Frozen final run, five seeds.

Method	min-class kept, keep 10%		worst-class mCA, keep 30%	
	CIFAR-10	CIFAR-100	CIFAR-10	CIFAR-100
Random	462.8	34.4	51.5	13.4
EL2N	138.6	4.8	47.4	1.3
GraNd	140.8	3.0	22.1	5.5
Forgetting	194.4	5.6	53.9	8.5
Entropy	179.2	4.2	50.4	5.6
CLIP-feat	168.0	4.0	50.8	8.4
DINOv2-feat	95.0	2.0	32.6	8.6

S9 Broader impact

Data pruning is adopted to cut the compute and storage of training. The finding here is that a coreset chosen for clean accuracy can carry a hidden robustness deficit, so a pruning decision taken purely on efficiency grounds may ship a model that is quietly more fragile under the shifted inputs it will meet in deployment. Reporting a shifted-test check alongside the clean number, as the main paper does, is a low-cost guard against that outcome.

References

1. Hendrycks, D., Dietterich, T.: Benchmarking neural network robustness to common corruptions and perturbations. In: Int. Conf. Learn. Represent. (ICLR) (2019)
2. Krizhevsky, A.: Learning multiple layers of features from tiny images. Tech. rep., University of Toronto (2009)
3. Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al.: DINOv2: Learning robust visual features without supervision. Trans. Mach. Learn. Res. (TMLR) (2024)

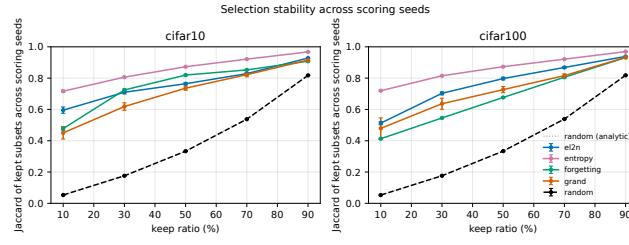


Fig. S4: Mean kept-set Jaccard across scoring-seed pairs against keep ratio. Score-based selection is more stable than random (dashed, analytic baseline). Frozen final run.

- Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Aspell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Int. Conf. Mach. Learn. (ICML) (2021)